

Davide Avagnano

Software Engineer · Full-Stack · TypeScript · Node.js · PostgreSQL

dav.avagnano@gmail.com | +39 331 4699097 | Naples, Italy · open to relocation / remote | linkedin.com/in/davide-avagnano | github.com/DavideAvagnano | portfolio

SUMMARY

Software Engineer with cross-functional full-stack skills and a **solid backend & architecture foundation**, in the **TypeScript / Node.js / React** ecosystem on PostgreSQL. I plan and model before writing code, apply common design patterns and best practices, and treat living documentation and scheduled refactoring as part of the definition of done. At home in the UNIX-like world (Linux, Bash, Docker), with hands-on SecOps and applied-AI work. Engineer by training and self-taught developer; I recently designed and shipped an entire SaaS **end-to-end**, from architecture to production.

SKILLS

Languages: TypeScript (advanced), JavaScript, SQL, Bash; Python (basics)

Frontend: React, Next.js (App Router / RSC), TypeScript, Tailwind CSS, shadcn/ui, state management (TanStack Query, Zustand), i18n

Backend & architecture: Node.js, Express, tRPC (type-safe APIs), REST, Zod; service-oriented & event-driven design, distributed workers (BullMQ, PostgreSQL LISTEN/NOTIFY), design patterns, domain modeling

Data & persistence: PostgreSQL (advanced), MySQL, Redis, Elasticsearch, Drizzle ORM; config-driven ETL & cross-platform data reconciliation

DevOps & UNIX: Linux (Ubuntu), Bash scripting, Docker, Nginx, PM2, CI/CD (GitHub Actions), Git/GitHub; SecOps (Suricata IDS, ELK, UFW, SSH)

Security & AI: PII encryption (AES-256-GCM), granular RBAC; OpenAI API, RAG (hybrid BM25 + kNN), applied LLM with cost governance

EXPERIENCE

Software Engineer — Freelance / Consultant

2026 – Present

Baaarber — multi-tenant SaaS platform, end-to-end product ownership for a client

- Designed and built **almost single-handedly** a multi-tenant SaaS (Next.js, React, tRPC, Drizzle ORM, PostgreSQL, Redis): ~57k LOC TypeScript, 31 tables, ~145 type-safe procedures — planning and modeling before coding, applying design patterns across the whole stack.
- Worked cross-functionally: PostgreSQL data modeling and type-safe backend logic through to the React UI; **event-driven workers** (BullMQ/PM2) with an outbox pattern, idempotent retries and distributed locking.
- Kept a **living-documentation approach** (50+ technical & functional docs, high-level comments) and scheduled refactors; enforced boundaries in CI (PII encryption AES-256-GCM, granular RBAC).
- Shipped an **applied-AI layer** (LLM assistant + translations) with cost governance; deployed on Linux (Nginx, Docker-ready, CI/CD via GitHub Actions).

Software Engineer — Freelance / Consultant

2025 – 2026

High-volume production platforms (ixily, Scalability, Hypefill) & SecOps tooling

- Analysed and improved complex production code** on marketing-intelligence platforms (Node.js/TypeScript, PostgreSQL, Redis, Elasticsearch): config-driven ETL, cross-platform data reconciliation and distributed workers (15+ PM2 processes).
- Built a service-oriented backend with **20+ external API integrations** (Meta Graph, Shopify GraphQL, analytics, call tracking) via reusable connectors and clean abstractions.
- Contributed to a **SecOps automation tool** (TypeScript + Bash): Suricata IDS + Elasticsearch + UFW firewall, remote rule lifecycle over SSH, ELK stack — hands-on UNIX / DevOps and security.
- Collaborated in **small teams** (3–5 devs) with CTO, design and marketing, in requirements calls, supporting colleagues on technical problems.

Software Development — Self-taught transition

2021 – 2024

Career switch from engineering into software: built web apps with React/Next.js on public APIs and strengthened algorithms, data structures and problem-solving, plus data analysis in Python.

EDUCATION

B.Sc. in Aerospace Engineering — University of Naples Federico II

2020 · 105/110

LANGUAGES

Italian (native) · English **B2** — professional working proficiency; daily reading and writing across codebases and technical documentation.